Code No: 152AG **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year II Semester Examinations, July/ August - 2021
ENGINEERING GRAPHICS
(Common to CSE, ITE)

Time: 3 Hours **Max. Marks: 75**

Answer any three questions
All questions carry equal marks

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1.a) Construct a diagonal scale of RF = 1/32 showing yards, feet and inches and to measure up to 4 yards.

b) Construct an ellipse when the distance of the focus from the directrix is equal to 80 mm and eccentricity is 3/5. [10+15]

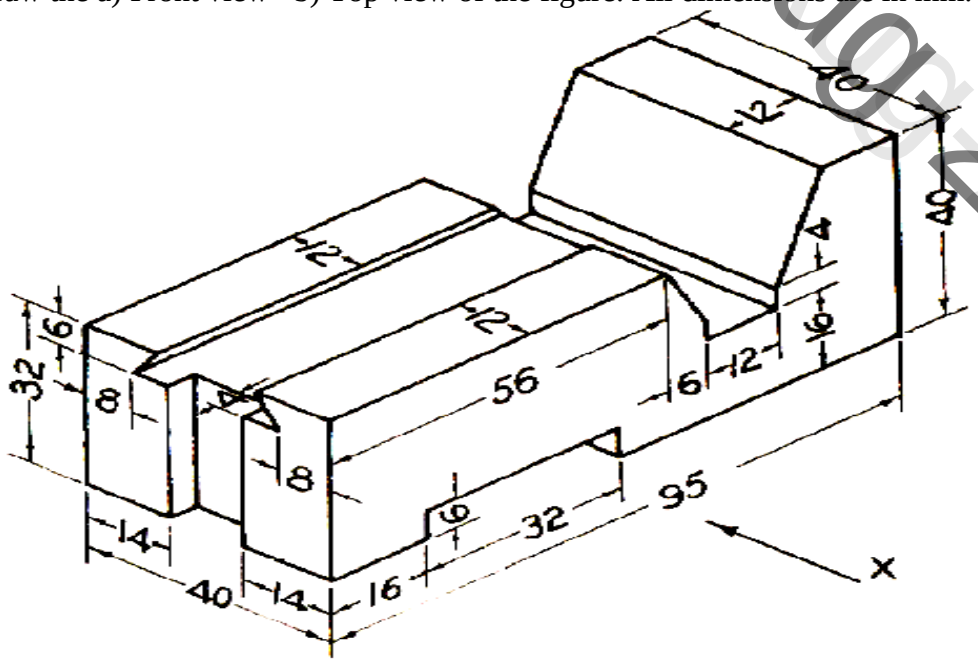
2. Draw the projections of a line AB, 90 mm long, its midpoint M being 50 mm above the HP and 40 mm in front of the VP. The end A is 20 mm above the HP and 10 mm in front of the VP. Find inclinations of the line with the HP and VP. [25]

3. A hexagonal pyramid, base 25 mm side and axis 50 mm long, has an edge of its base on the ground. Its axis is inclined at 30^0 to the ground and parallel to the VP. Draw its projections. [25]

4. A conical shape vertical chimney of 60 m base diameter joins a roof sloping at an angle of 35^0 with the horizontal. The shortest portion over the roof is 25 m. Determine the shape of the sheet metal from which the chimney can be fabricated. Take appropriate scale. [25]

5. Draw an equilateral triangle of 75 mm side and inscribe a circle in it. Draw the projections of the figure, when its plane is vertical and inclined at 30^0 and one of the sides of the triangle is inclined at 45^0 to the HP. [25]

6. Draw the a) Front view b) Top view of the figure. All dimensions are in mm. [10+15]



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